

Shooting Range Level Design Document,

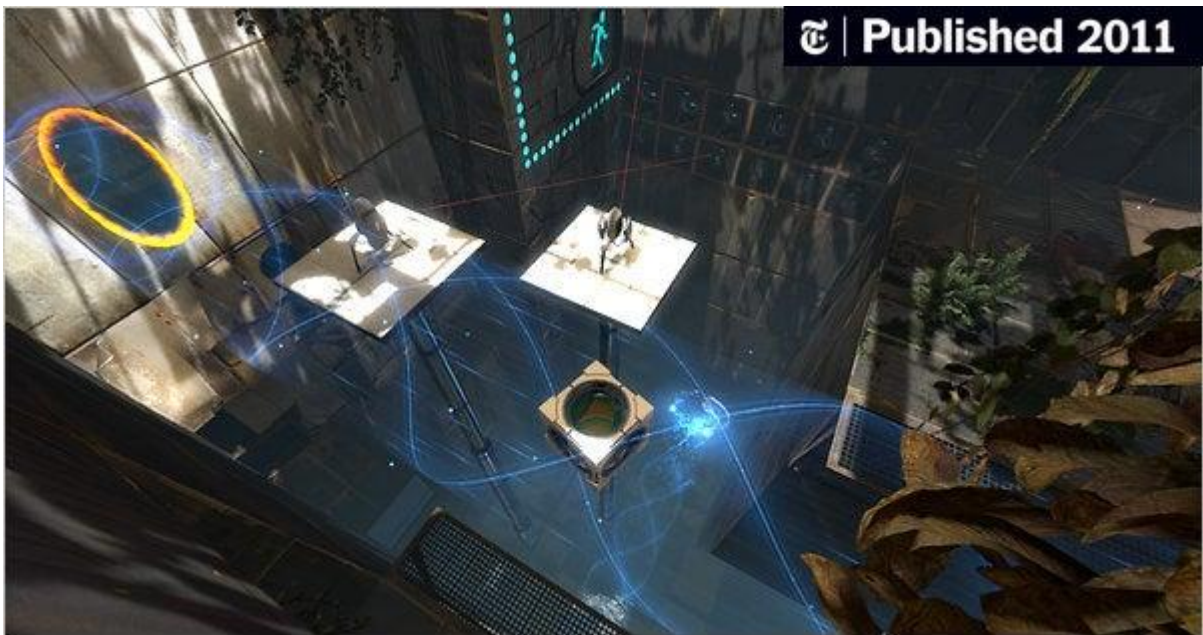
Design Brief And Technical Spec

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Game Level Concept and Intentions:

- The player is in a shooting range level
- They have multiple guns they can swap between
- Each gun has different functionality
- There are moving targets for the player to shoot
- The targets add to the player's score when shot
- The targets change colour with different colour targets worth different amounts of score
- Targets will lose colour and stop moving upon being hit to indicate they are deactivated
- Obstacles stand between the player and the targets
- Destroying an obstacle will lower the player's score
- The first part of the level is a warmup where the player can take their time
- The rest of the level the player must perform their best while on a moving platform before it reaches the end
- At the end of the level, the player is given their final score
- The player may reset the level by stepping onto a reset platform
- The level will also reset upon completion or upon the player entering a forbidden zone such as by falling off the platform
- The player may use the special shrink cannon to neutralise obstacles without losing score
- The course the platform takes the player changes form factor as the player progresses

Reference Images:



Level Purpose:

- For the player to test their marksmanship skills through the use of a variety of guns in a shooting range.
- Using their timing skills, players can try and shoot targets when they are worth the most points.
- Encouraging player experimentation with the different guns, as players use them to their advantage based on what would be best to use at the current moment.
- To discourage player losing score through destroying obstacles.
- For the player to be able to reset the level and try improve upon their previous score.

Challenges:

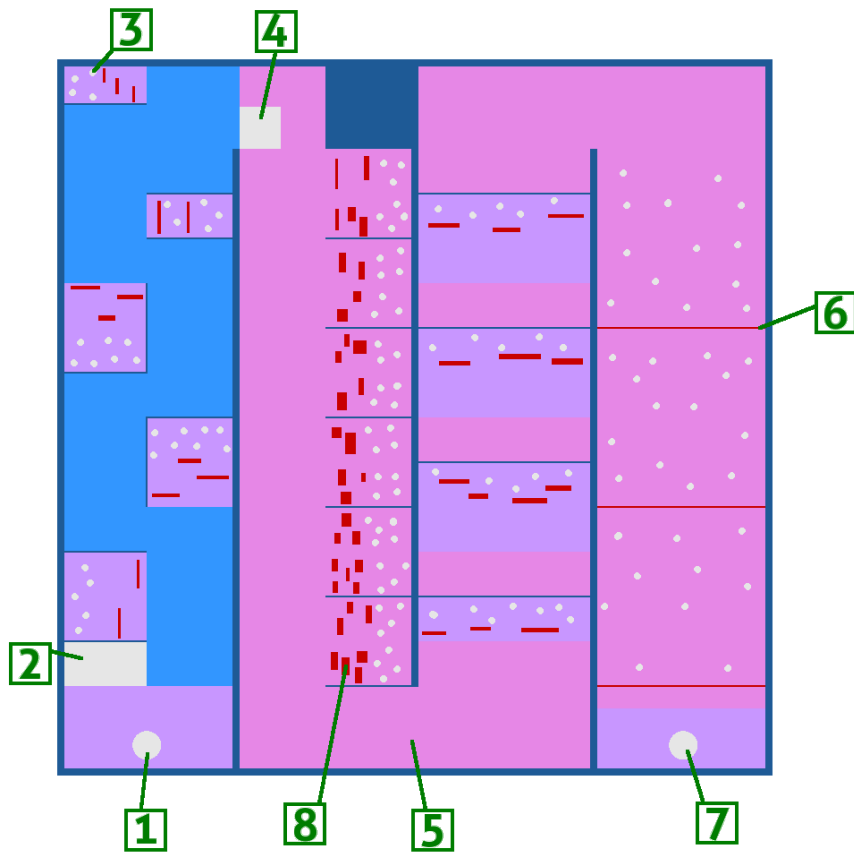
- Aiming
- Timing
- Avoiding Obstacles
- Using the right gun at the right time

Elements:

- Start point
 - Where the player begins the game
- Goal
 - Gives the player final score and resets the game
- Reset platform
 - Manually resets the game for the player
- Moving Platform
 - Carries the player through the level upon the player stepping on it
- Moving Obstacles
 - Impede the player's aim at targets, lowers score if destroyed by 10 points
- Forbidden zones

- Reset the game if the player falls off the platform/enters an area they shouldn't
- Score mechanic
 - Tracks the player's performance
- Moving targets
 - Change colour every few moments, give the player score depending on their colour. Green is worth the most points at 50, yellow is worth a moderate amount of points at 20 and orange is worth the least amount of points at 10
- Guns
 - Assault Rifle - Automatic weapon with high rate of fire, high accuracy, lower damage per bullet and moderate range
 - Shotgun - Single fire weapon with a low rate of fire, lower accuracy but fires multiple, higher damage bullets with a higher range
 - Shrink Cannon - Raycast beam weapon that shrinks certain targets but deals no damage. High rate, high accuracy but lowest range

Topdown Map:



1. Start Point
2. Reset Platform
3. Moving Targets
4. Moving Platform
5. Forbidden Zone
6. Barrier
7. Goal
8. Moving Obstacles

Technical Spec:

- Lose Condition:
 - Player enters forbidden zone
 - Player overlaps with trigger volume
 - Event is triggered that prints player "You lose!" message
 - Level is reset after a delay of a few seconds
- Win Condition:
 - Player reaches goal at end of level
 - Player overlaps with trigger volume
 - Event is triggered that prints player "You win!" message
 - Player score is received from player blueprint
 - Final score is printed
 - Level is reset after a delay of a few seconds
- Reset Platform:
 - Player steps on platform
 - Player overlaps with a trigger volume
 - Event is triggered that immediately resets the level
 - Has a reset sign above it
 - Is a large white platform
- Score:
 - Is contained with the player blueprint
 - Is updated whenever a target's health is depleted
 - Final score is given once player reaches goal
- Moving Targets:
 - Begin with a random colour/material
 - Each colour/material is assigned a score value in the target blueprint
 - Cycles between the colours in the order of green -> yellow -> orange
 - Green is worth 50 points
 - Yellow is worth 20 points
 - Orange is worth 10 points
 - Has light

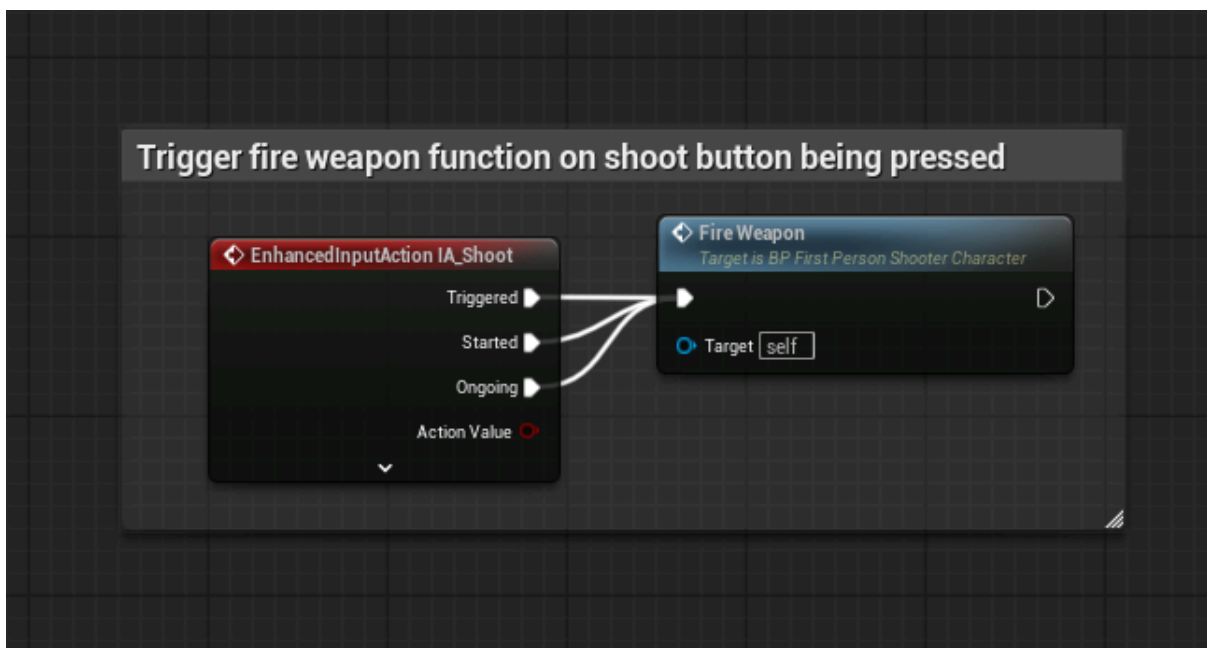
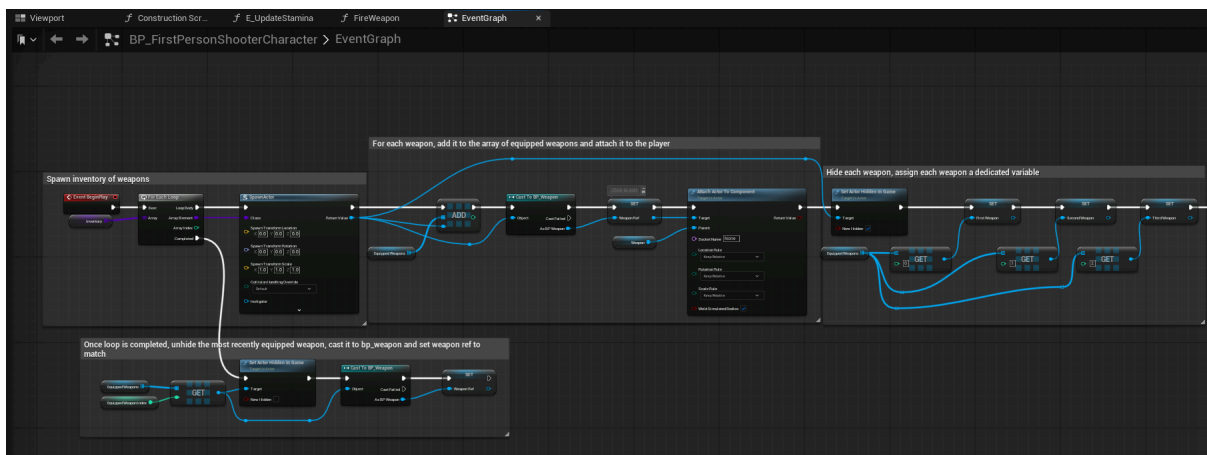
- Is a flat cylinder
- Moves in a cycle that syncs with their colour changes
- Has a health component
- Once health is depleted updates score to the player blueprint as appropriate, loses colour (becomes grey), loses light and stops movement
- Moving Obstacles:
 - Prevent the player from shooting targets
 - Move up and down or side to side
 - Have a health component
 - On health being depleted, is destroyed and subtracts 10 score from the player blueprint
 - Can be shrunk by the shrink cannon
 - Bright red to indicate they should be avoided
- Barrier:
 - Blocks the player's path before the goal
 - Can be destroyed by depleting health or shrunk
 - Doesn't effect score
 - Will cause the player to fall if not dealt with
- Start Point:
 - Where the player starts the level
- Goal:
 - Objective point at the end of the level.
- Moving platform:
 - Bright white platform
 - Has an InterpToMovement component
 - InterpToMovement event tick is paused on EventBeginPlay
 - Has a trigger volume
 - Waits for the player to overlap with the trigger volume
 - Once the trigger volume is overlapped, the event tick resumes on InterpToMovement
 - The platform carries the player through the rest of the course

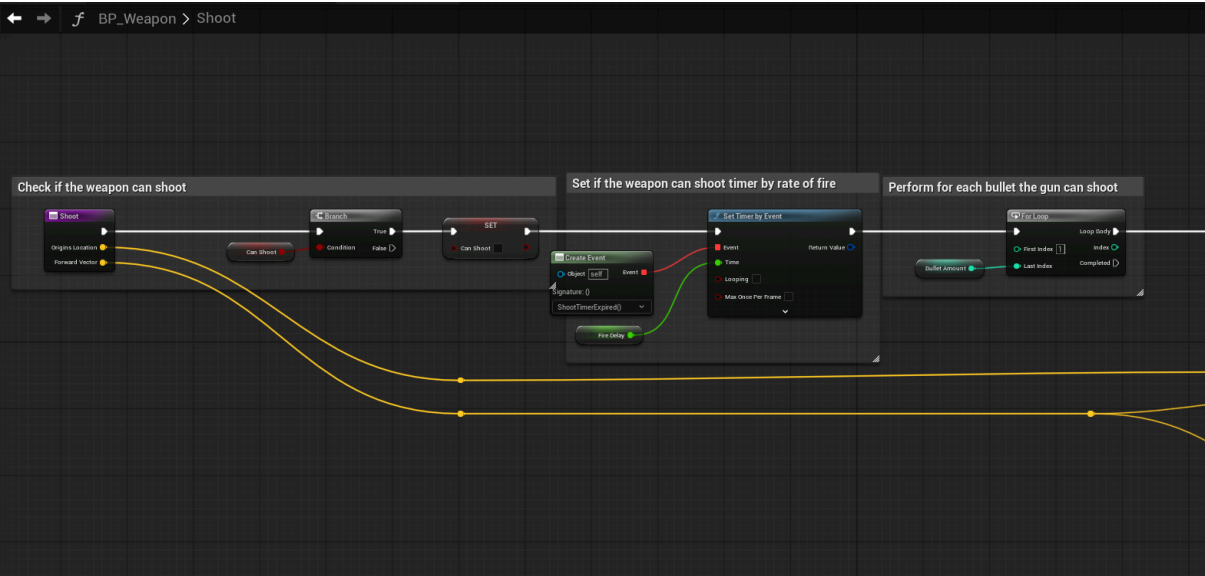
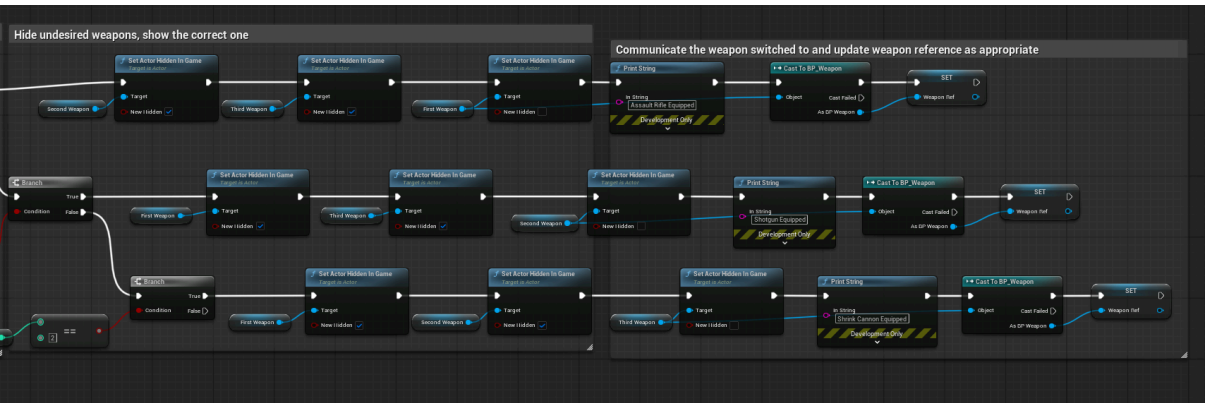
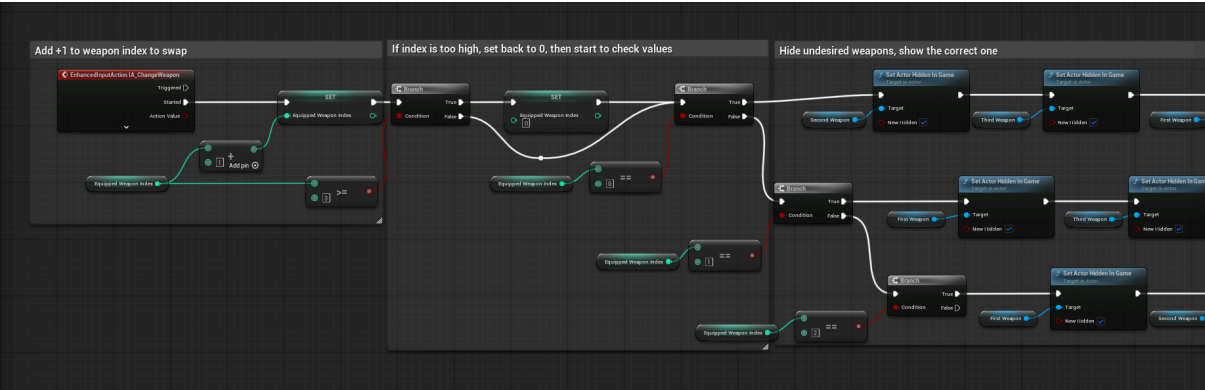
- The player has until the platform completes its course to hit as many targets with as high of a score as they can
- Once the platform reaches the end, it becomes unmovable so it cannot be shrank
- A barrier spawns on the back of the platform to prevent the player shooting targets behind them
- Weapons:
 - BP_Weapon is the parent weapon blueprint
 - Contains raycasting code within shoot function
 - Contains bullet deviance function
 - Contains multiple variables inherited by child weapon blueprints;
 - Damage (Float)
 - Fire Delay (Float)
 - Bullet Accuracy (Float)
 - Range (Float)
 - CanShoot (Boolean)
 - Bullet Amount (Integer)
 - Shrink Amount (Vector) [For Shrink Cannon]
 - Is Shrink Gun (Boolean) [For Shrink Cannon]
 - BP_WeaponAssaultRifle:
 - Is a child of BP_Weapon
 - Has low damage
 - Has very low fire delay (automatic weapon)
 - Has high bullet accuracy
 - Has moderate range
 - Shoots a single bullet at a time but in rapid succession
 - BP_WeaponShotgun:
 - Is a child of BP_Weapon
 - Has high damage
 - Has a moderate fire delay (single fire weapon)
 - Has lower accuracy (spreadshot)
 - Has higher range

- Shoots multiple bullets in a spread
- BP_ShrinkCannon:
 - Is a child of BP_Weapon
 - Has no damage
 - Has a low fire delay (beam weapon)
 - Has high accuracy
 - Has low range
 - Shrinks an actor by multiplying its scale by 0.9 of its previous value
- Player:
 - Named BP_FirstPersonShooterCharacter
 - Can sprint (previously implemented)
 - Has stamina (previously implemented)
 - Contains score variable (previously implemented)
 - Contains inventory array that holds weapons
 - Changes weapon on button press (Right click)
 - Puts inventory of weapons into equipment array on EventBeginPlay
 - Contains variables for weapons and other mechanics;
 - Sprint Walk Speed (Float)
 - Normal Walk Speed (Float)
 - Stamina (Float)
 - Can Gain Stamina (Boolean)
 - Timer Active (Boolean)
 - Max Stamina (Float)
 - Max Score (Integer)
 - Player Score (Integer)
 - Health (Float) [From Health Component]
 - Inventory (Class Array)
 - Equipped Weapons (Actor Array)
 - Equipped Weapon Index (Integer)
 - First Weapon (Actor)
 - Second Weapon (Actor)
 - Third Weapon (Actor)
- Health Component

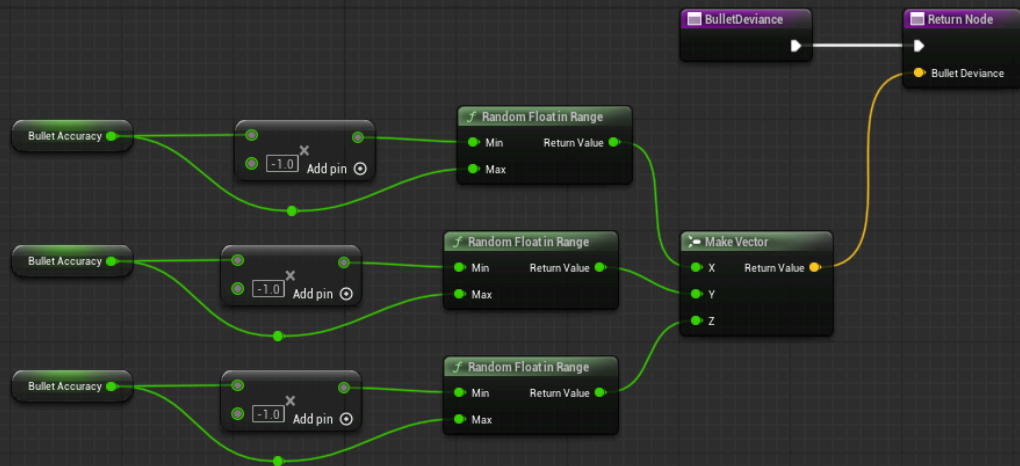
- Contains the Health variable
- Contains events OnHealthChanged and OnHealthDepleted
- Used for target and obstacle blueprints
- Level Blueprint:
 - Contains events for resetting the level

Blueprint Code:



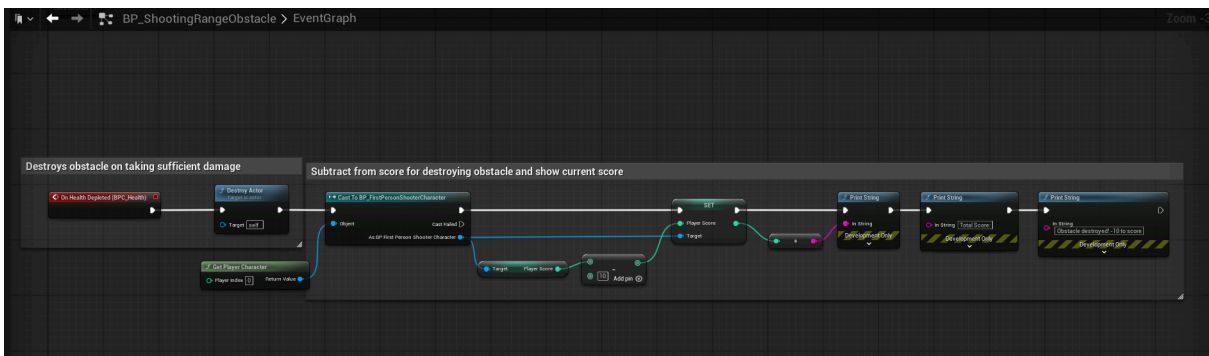
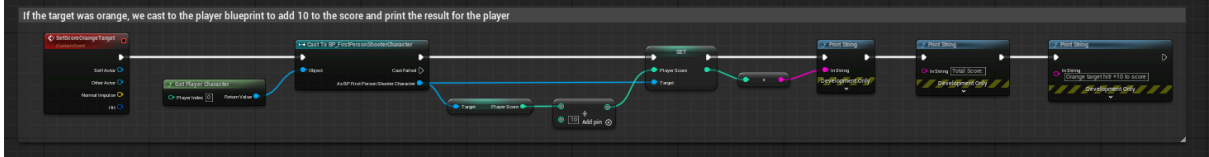
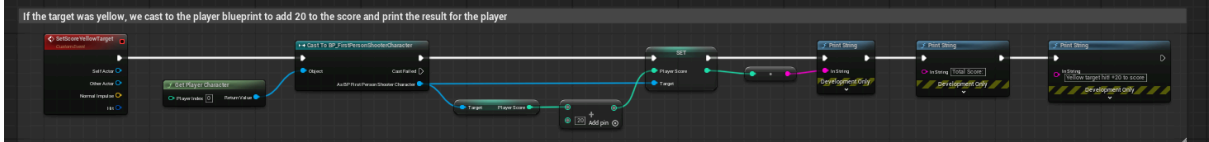
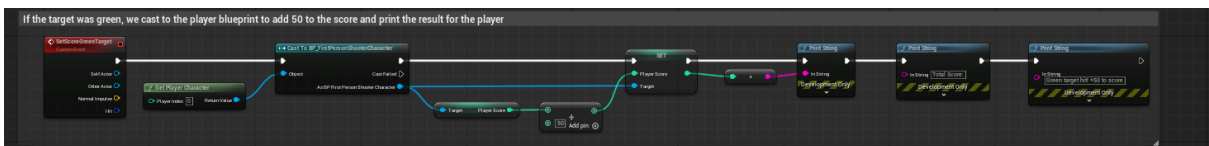
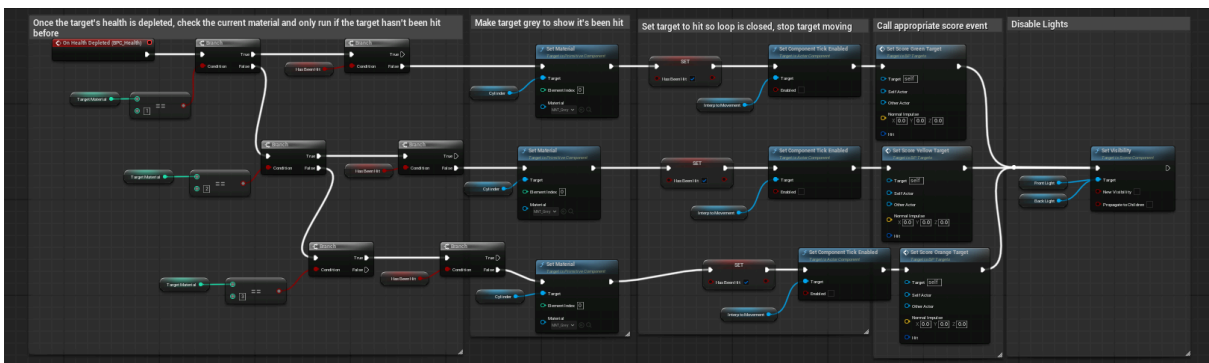
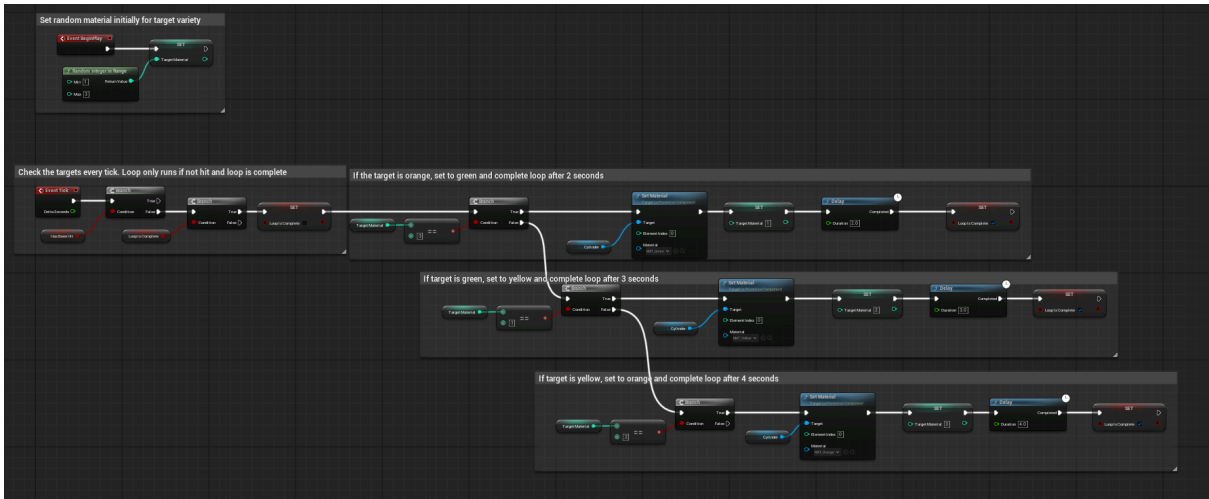


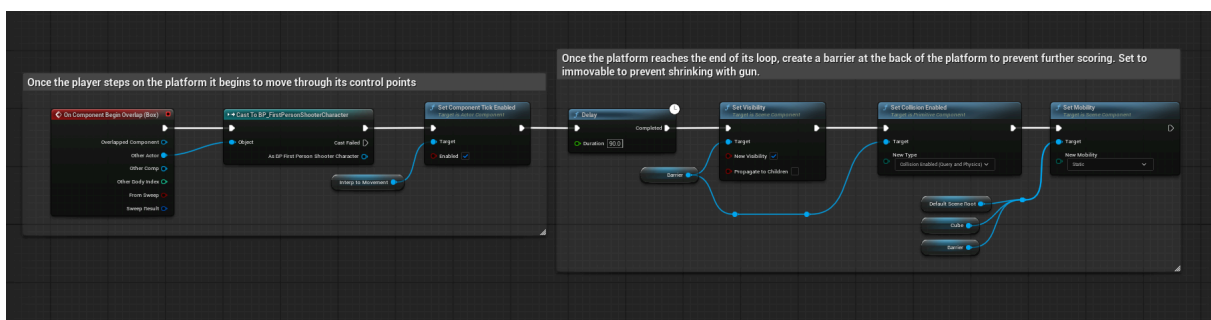
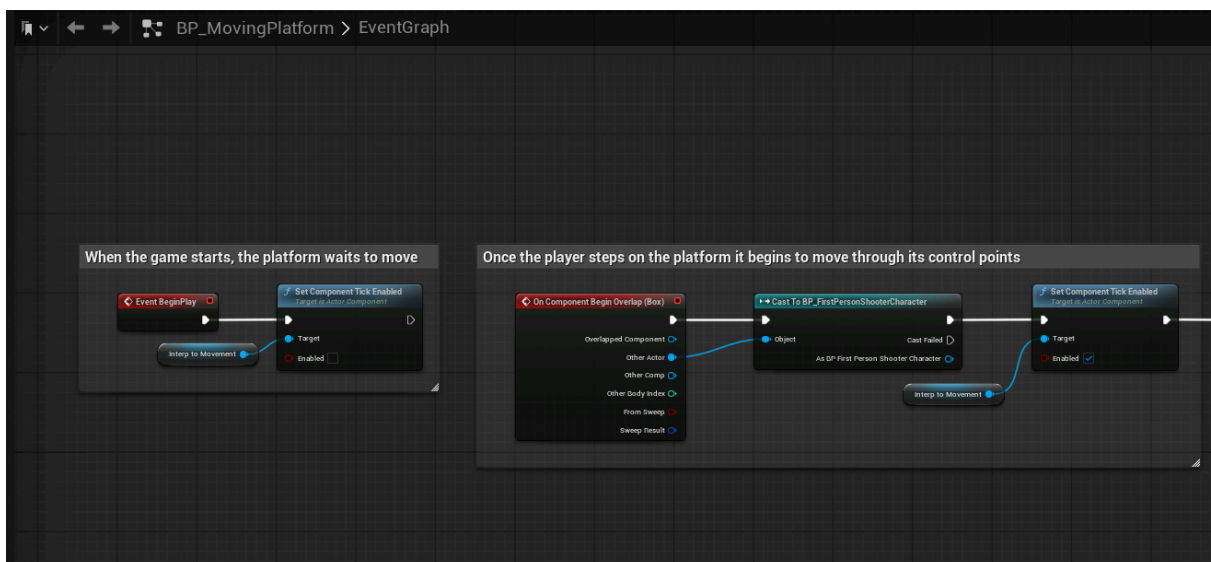
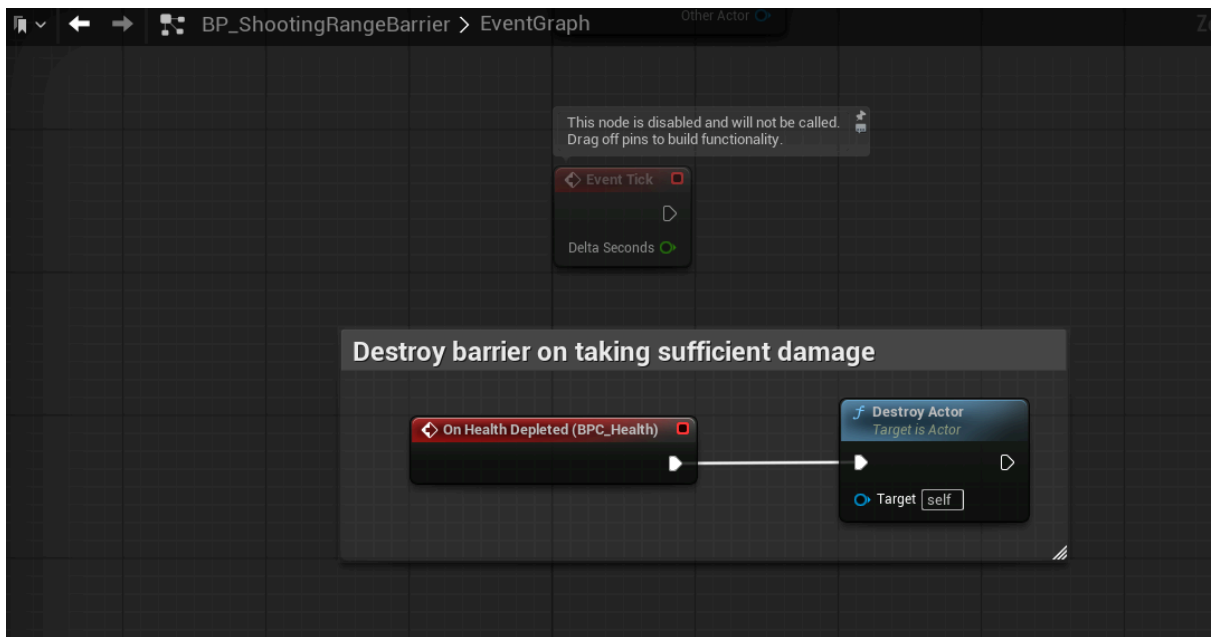
Multiply gun bullet accuracy by -1 for minimum value, use default value for maximum value. Do this three times to make a vector and set that as bullet deviance.

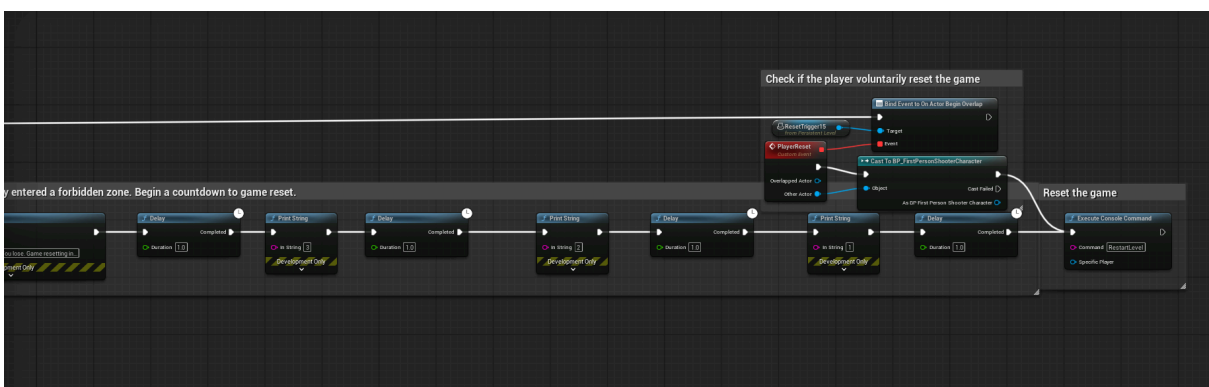
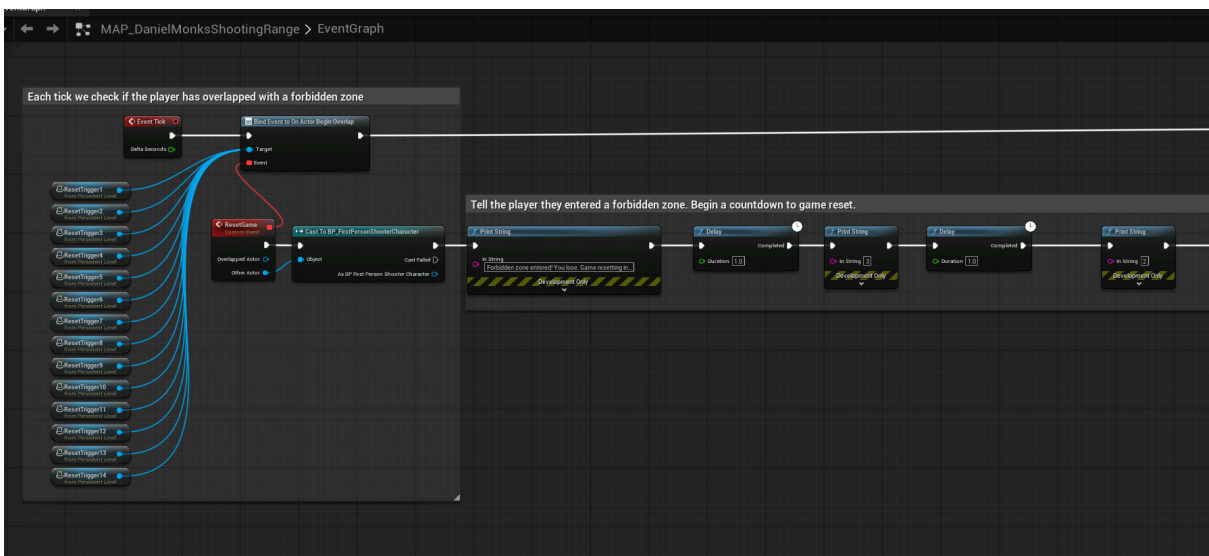
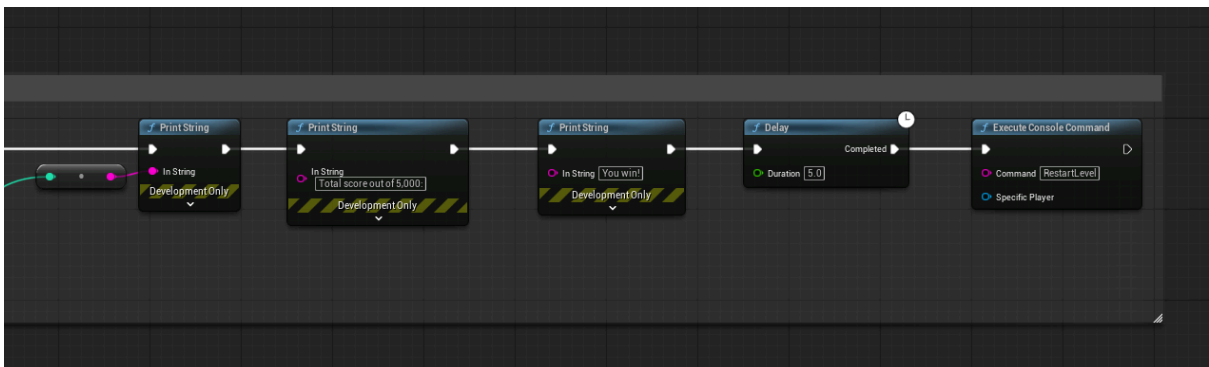
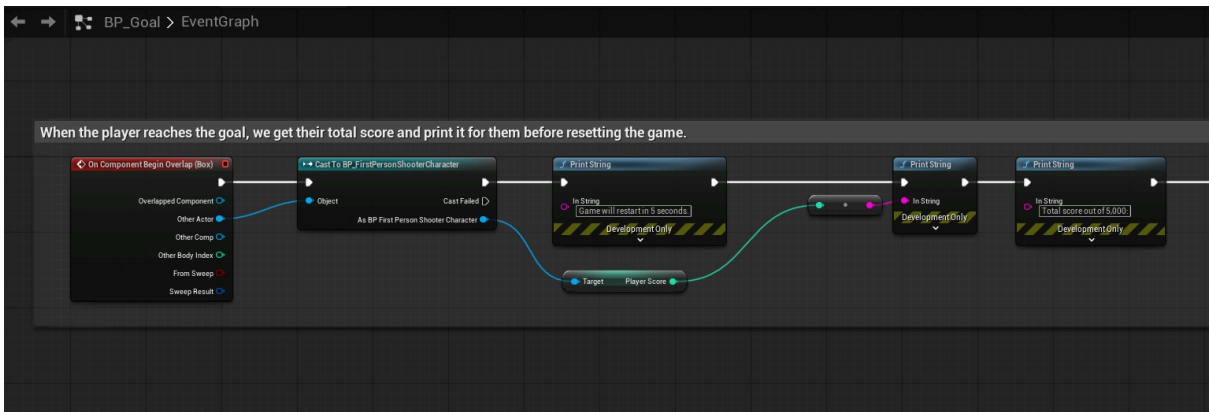


Only let the gun shoot if its timer is expired, determined by rate of fire in the shoot function









Tutorials Used:

- [How to Make a Moving Platform in Unreal Engine 5](#)
- [Creating a Gun Blueprint & Setting up a Simple Inventory | Unreal Engine 5 FPS Game Tutorial #16](#)
- [\[Unreal Engine 5\] How to setup a Shotgun LINETRACE](#)